

IVs and DVs, oh my!

DV: Dependent variable (a.k.a. outcome variable): The variable you are trying to explain.

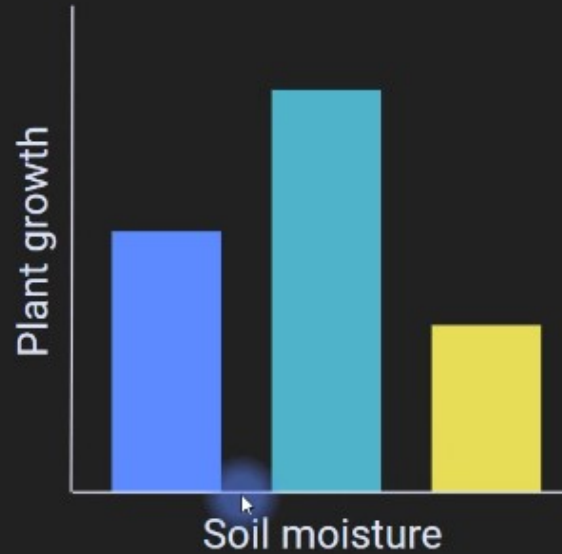
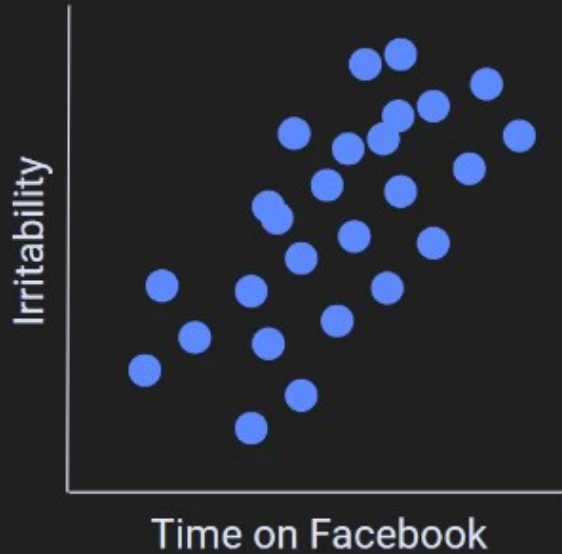
IV: Independent variables (a.k.a. explanatory variables): The variables that you hope will explain the DV.

Examples:

- Effects of soil moisture on plant growth.
- Effects of time spent on Facebook on irritability.
- Relation between money spent on clothes and bar-hopping.

IVs and DVs, oh my!

Usually, the IV is on the x-axis and the DV is on the y-axis.



Model-fitting in statistics

A model is an equation that explains some features in a dataset.

$$h = \beta_1 s + \beta_2 p + \beta_3 n + \varepsilon$$

h = height

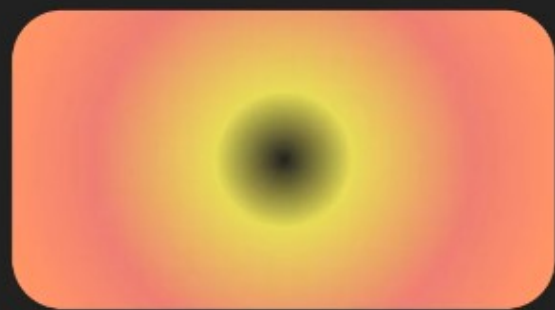
s = sex (m/f)

p = parents' height

n = childhood nutrition

ε = residual

Residuals



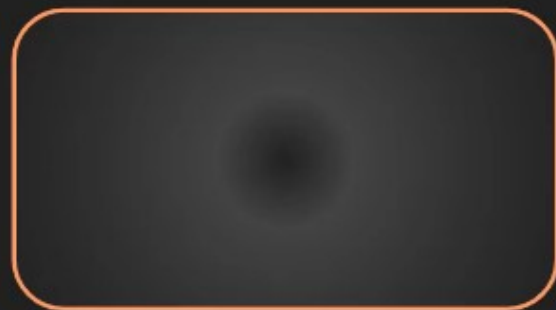
Data (real world)

-



Model

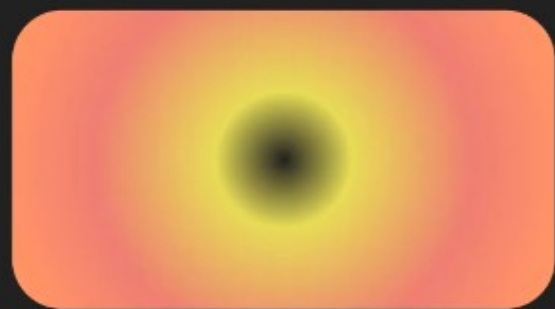
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Residual (error)

Important point:
Residuals should be small,
but models should be simple!

Residuals



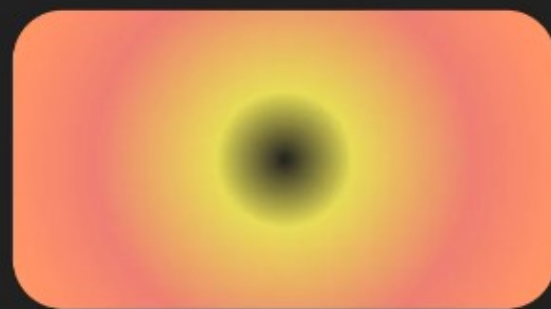
Data (real world)

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Model

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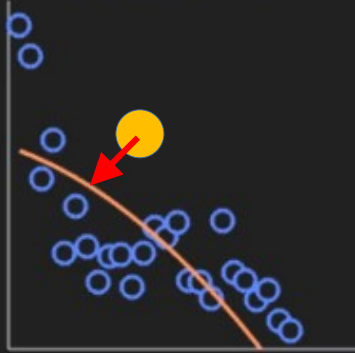


Residual (error)

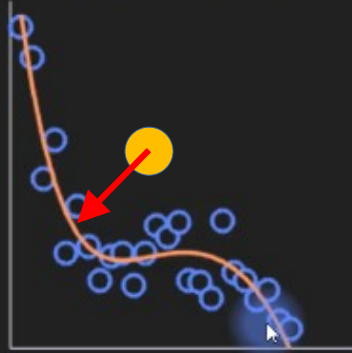
Important point:
Residuals should be small,
but models should be relevant!

How complex should model be?

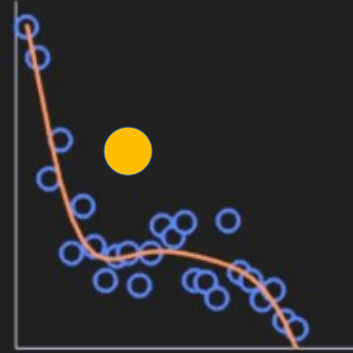
Order = 2



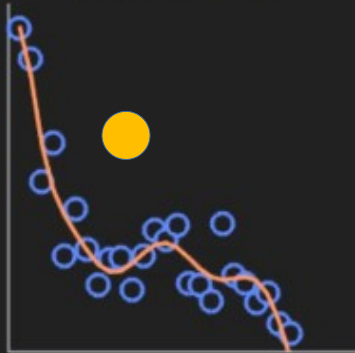
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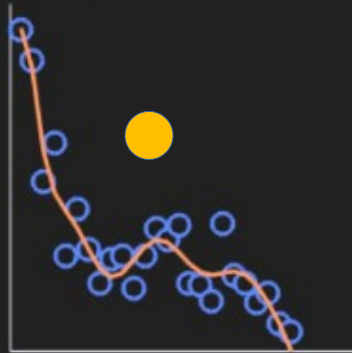
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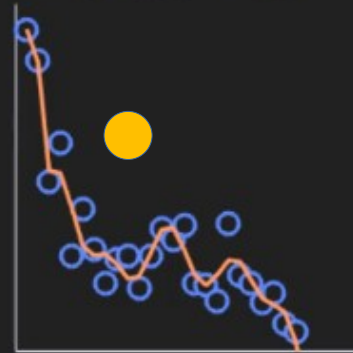
Order = 11



Order = 14



Order = 17



Model
karmaşıktıkça,
yeni gelen
datanın bu
modele uyma
ihtimali azalıyor !

(BIAS/VARIANCE
problem :))

Hypotheses and models

Important point:

All hypotheses are model comparisons!



is better than

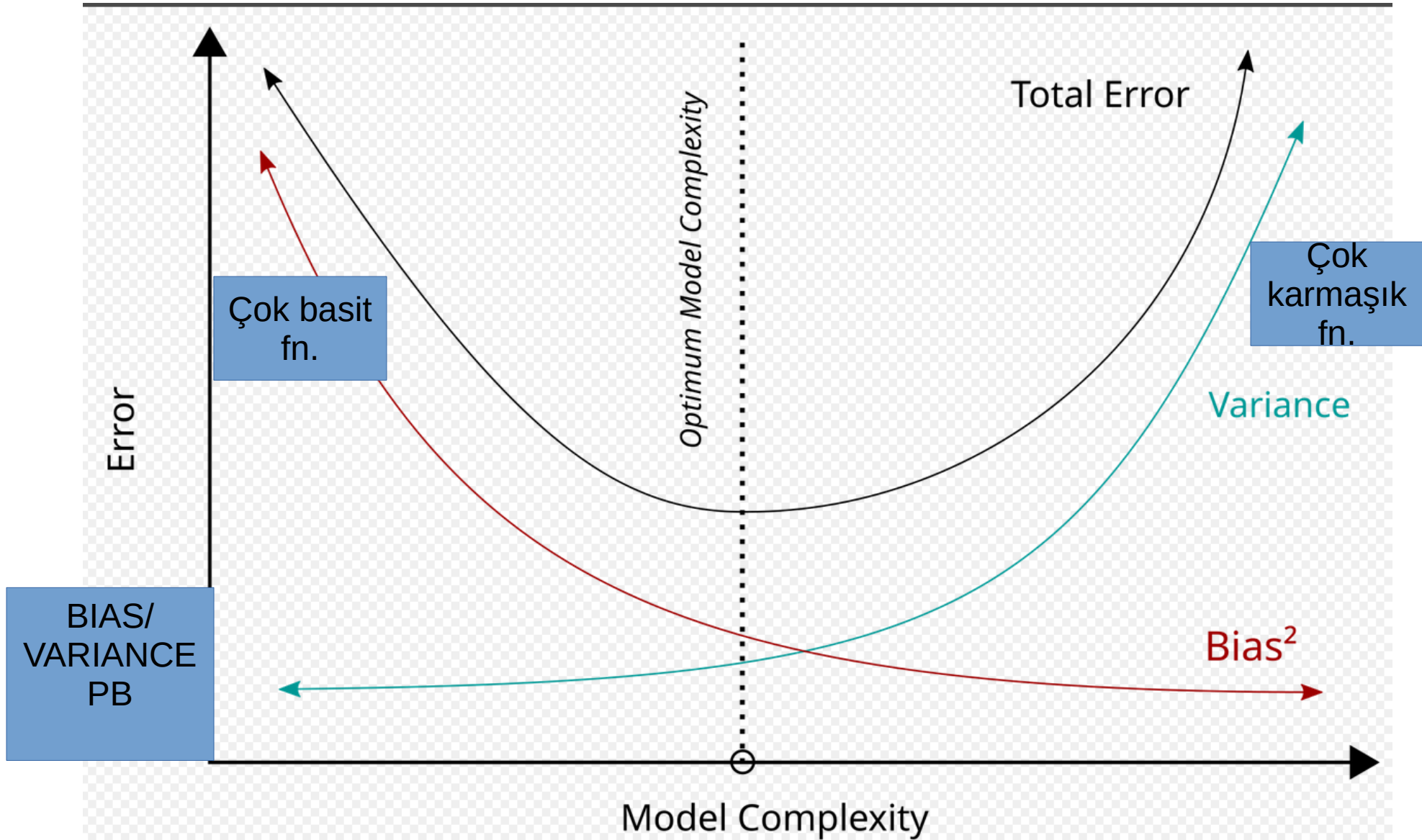


How complex should model be?

Take-home message:

A model should be as simple as it can be,
and as complicated as it must be.

BIAS / VARIANCE PROBLEM
UNDER-FITTING / OVER-FITTING PROBLEM



What are hypotheses and how do you specify one?

In this video, you will learn

- Why hypotheses are so important in science.
- What it means for a statement to be an “hypothesis.”
- Examples of strong (good) and weak (bad) hypotheses.
- What the “null hypothesis” means.

What is an hypothesis?

A falsifiable claim that requires verification, typically from experimental or observational data, and that allows for predictions about future observations.

* Different fields have somewhat different definitions.

Why are hypotheses important?

Hypotheses improve experiment design, critical thinking, and data analyses.

Hypotheses transform loose ideas into concrete and specific claims.

Hypotheses are used to develop new and more accurate theories, and to dissolve bad theories.

Most progress in science, engineering, and medicine is the result of hypothesis-testing.

What makes a strong hypothesis?

A strong hypothesis is:

- Clear
- Specific
- Falsifiable
- Based on prior data or theory
- Leads to a statistical test
- A statement, not a question
- A prediction about the direction of an effect.
- Relevant for unobserved data or phenomena
- Relevant for understanding nature

Examples of hypotheses

Not an hypothesis:

Medical research is important for curing diseases.

Weak but still an hypothesis:

The medication has an effect.

Strong:

The medication reduces symptom X in a dose-response fashion.

Examples of hypotheses

Not an hypothesis:

Will students pass this course?

Weak but still an hypothesis:

Studying improves grades.

Strong:

A combination of self-study and group-study will improve final exam grades by at least 10%.

Examples of hypotheses

- Not** 1. There are other universes with different physical rules.
- Strong** 2. Wearing purple underwear improves mood.
- Weak** 3. Plants grow differently in sugar-water.
- Not** 4. Mike X Cohen's courses are awesome.
- Strong** 5. Washing hands for 20 seconds reduces disease spread.
- Weak** 6. An apple a day keeps the doctor away.
- Not** 7. Are people more creative after watching stand-up comedy?

The null hypothesis

The “null hypothesis” is the hypothesis that nothing interesting is happening in the data.

In the research, you specify the “alternative hypothesis” (should be called the “effect hypothesis”).

In statistical analysis, you test the null hypothesis.

Often referred to as H_0 (“H nought”)

The null hypothesis: example

H_a : People will buy more widgets after seeing advertisement X compared to advertisement Y.

H_0 : The advertisement type has no effect on widget purchases.

Proving hypotheses?

It is not possible to *prove* an hypothesis.

Hypotheses can be rejected or fail to be rejected
(interpreted as being supported).

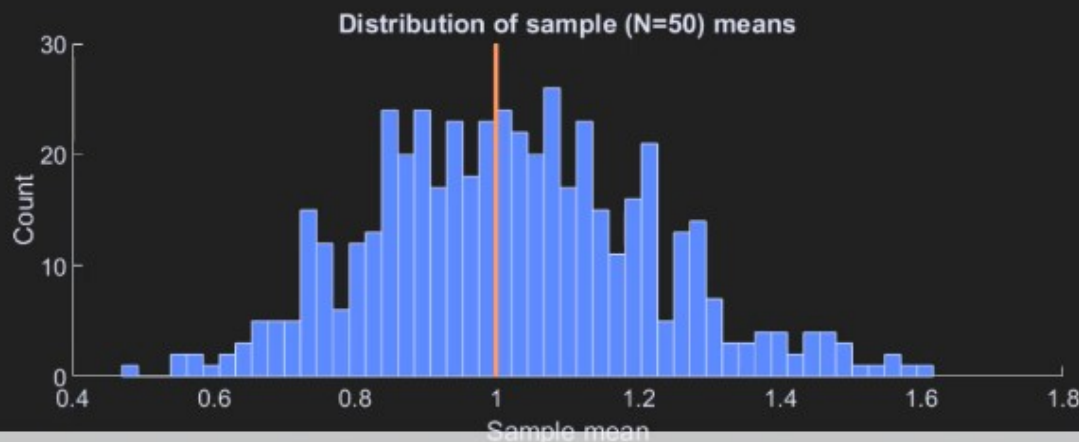
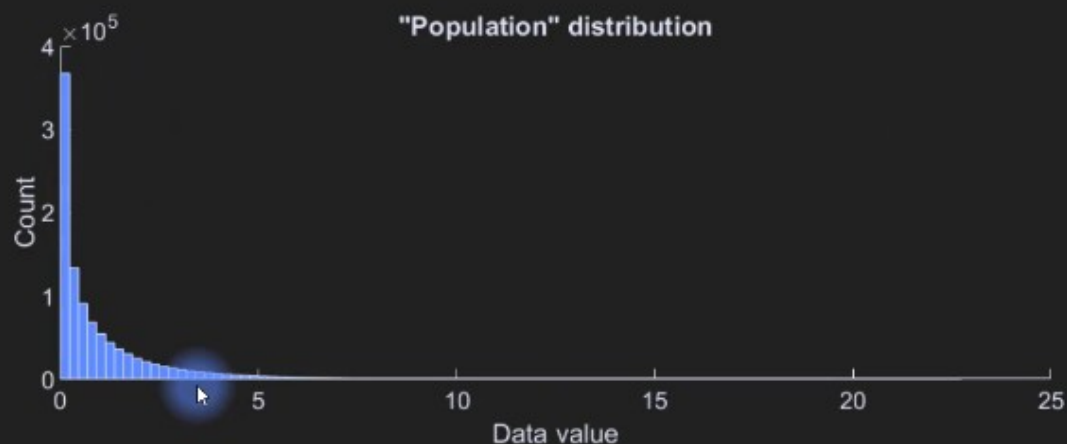
Distributions under the null and alternative hypotheses

In this video, you will learn

- More about sample estimates distributions.
- How sampling variability and noise contribute to hypothesis distributions.
- THE BASIS OF ALL INFERENTIAL STATISTICS.

Reminder about distributions of estimates

(From lecture on CLT.)



Creating a null hypothesis distribution

Scientific question: Are giraffes taller than giraffes?



vs.



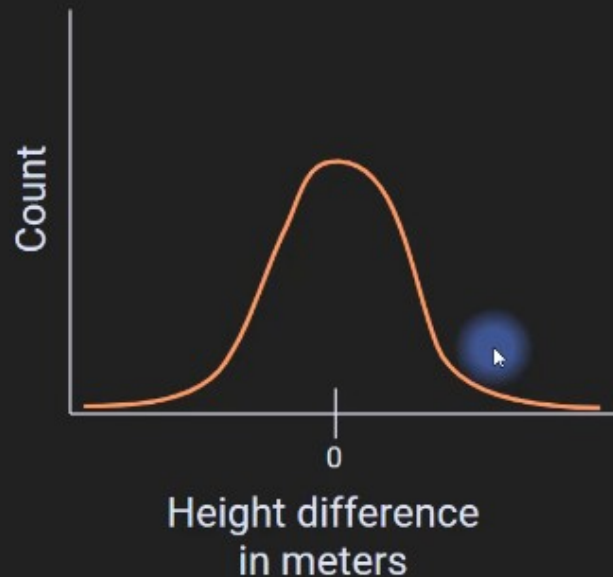
Creating a null hypothesis distribution

Scientific question: Are giraffes taller than giraffes?

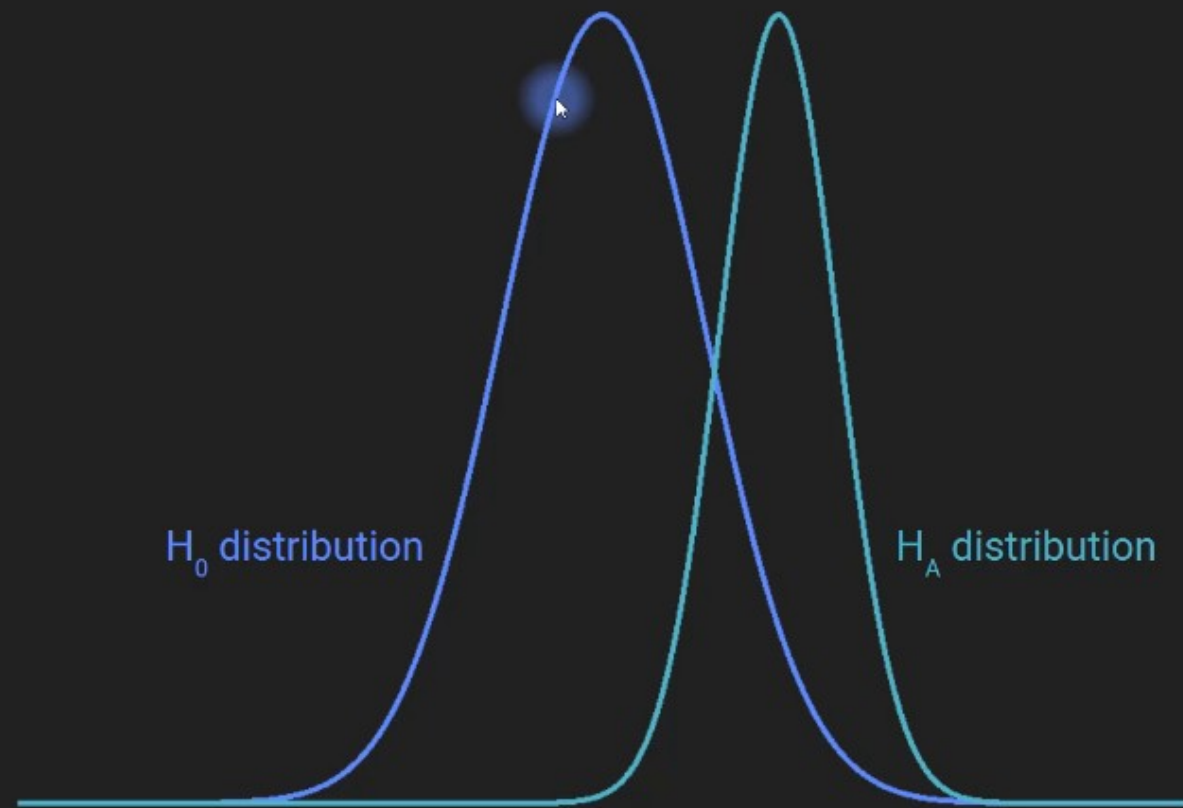


Creating a sample distribution of differences

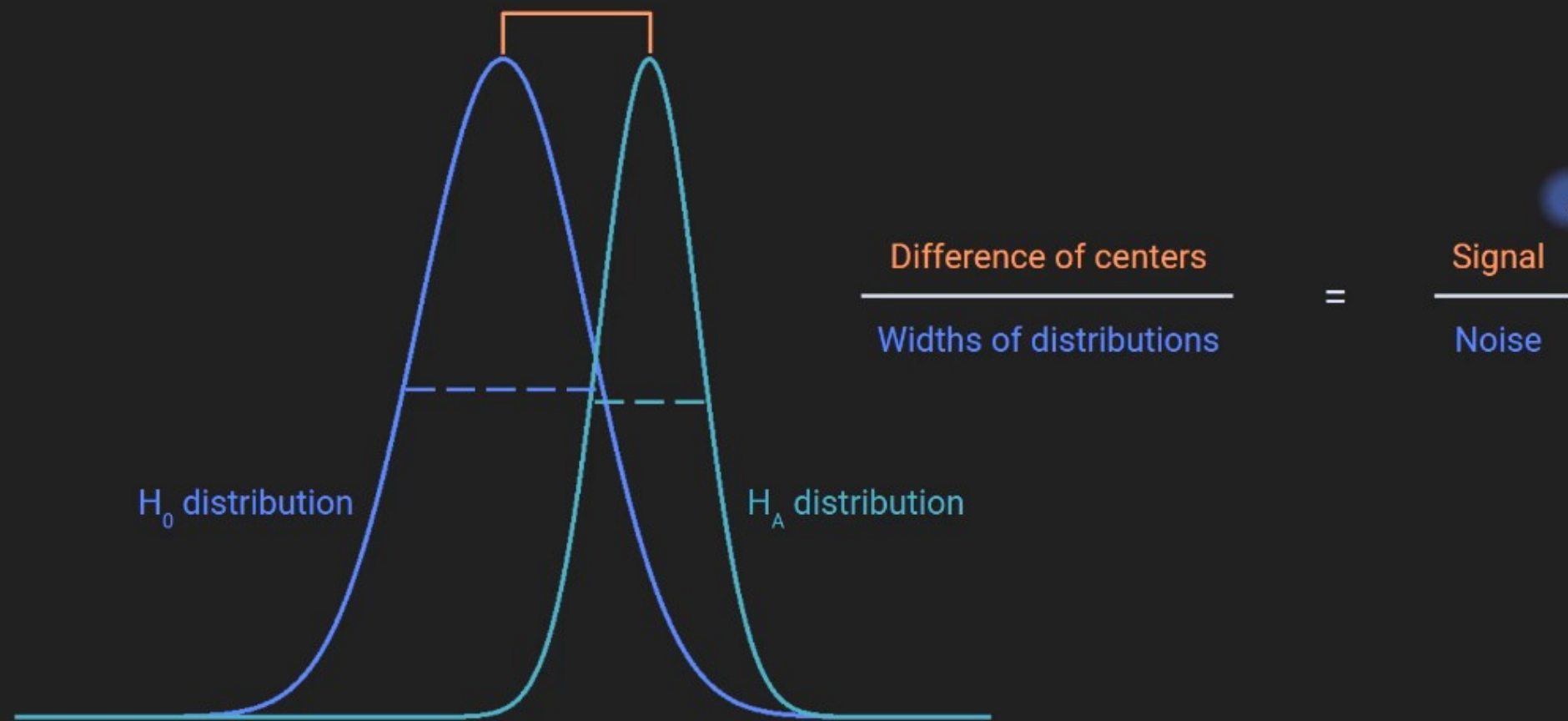
Scientific question: Are giraffes taller than giraffes?



Estimate distributions under H_0 and H_A



The basis of ALL inferential statistics



P-values: definition, tails, misinterpretations

In this video, you will learn

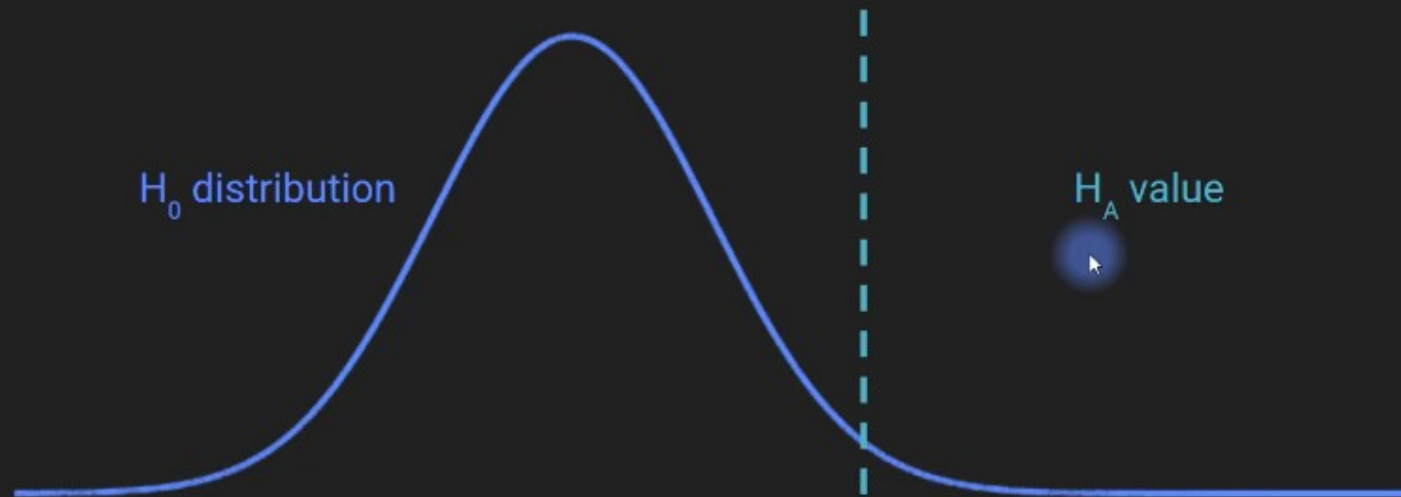
- Where p-values come from and what they mean.
- What the “tails” are and how they affect p-values.
- Common misinterpretations of p-values to avoid.

What is a p-value?



Note: These are distributions of sample parameter estimates, not distributions of the data!

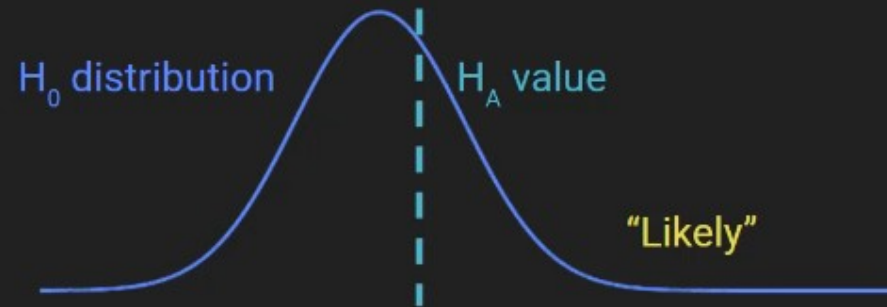
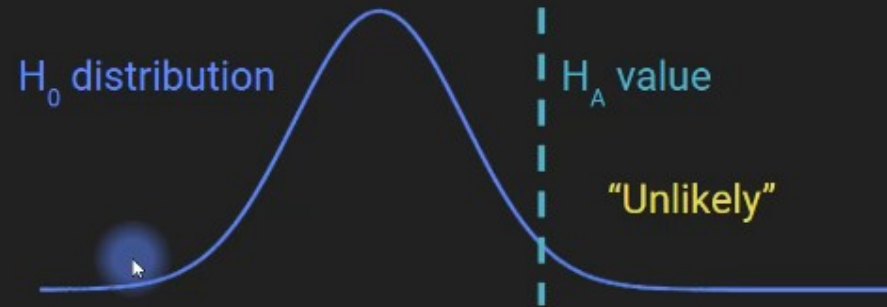
What is a p-value?



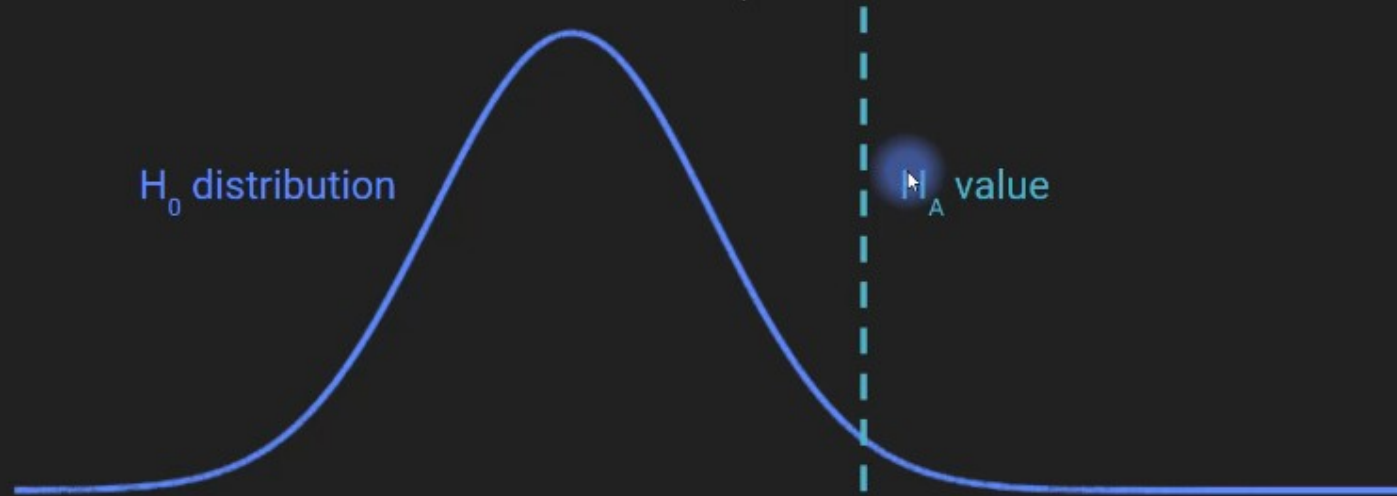
What is a p-value?

Questions:

- How likely is the H_A value to occur if H_0 is true?
- What is the probability of observing a parameter estimates of H_A or larger, given that there is no true effect?
- What is $P(H_A | H_0)$?



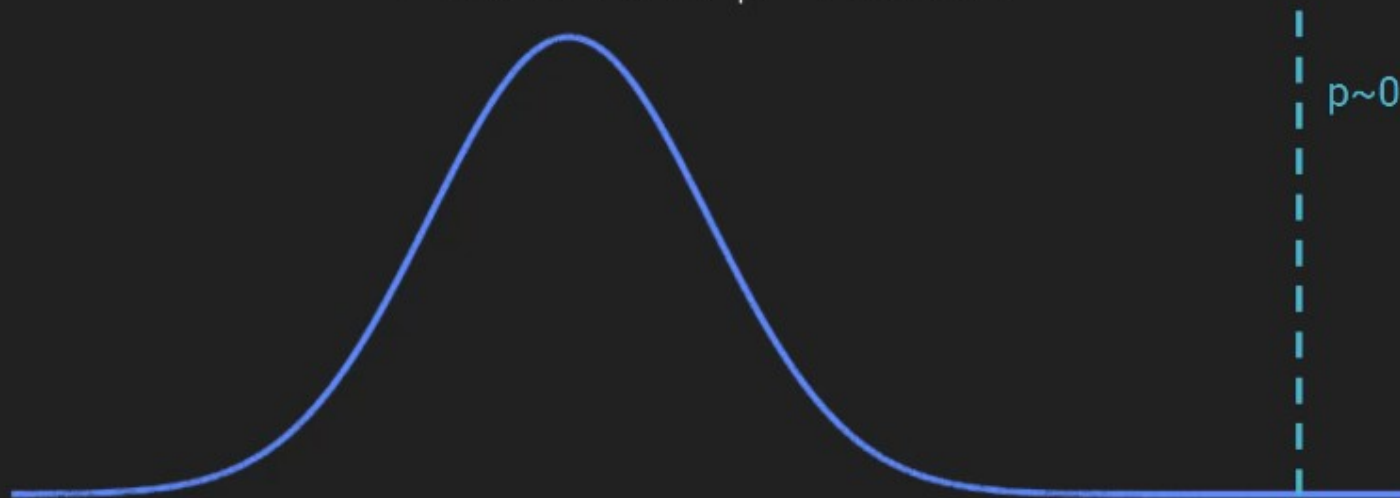
What is a p-value?



Important concept:

We cannot prove that H_A is true. We can only compute the probability that the test statistic associated with H_A could be observed given that there is no true effect.

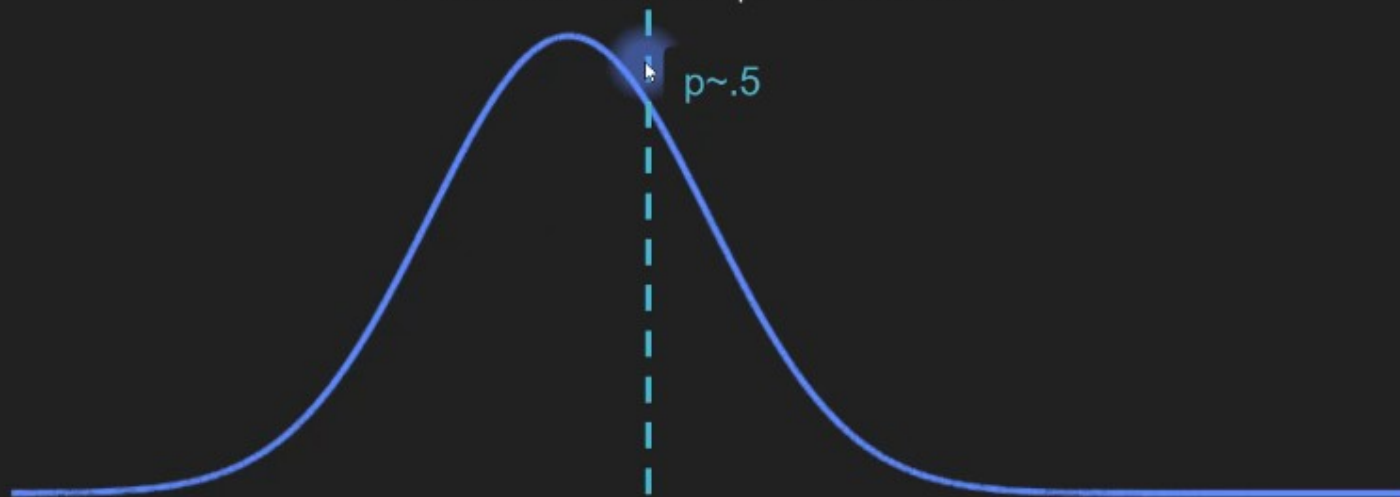
What is a p-value?



P-values are probabilities. They range from 0 to 1.

Values closer to zero indicate low probability of $H_A | H_0$, and values closer to one indicate high probability of $H_A | H_0$.

What is a p-value?



P-values are probabilities. They range from 0 to 1.

Values closer to zero indicate low probability of $H_A | H_0$, and values closer to one indicate high probability of $H_A | H_0$.

When is a finding “statistically significant”?



A finding is called “statistically significant” if the test statistic is greater than a threshold. That is, if $p(H_A) < p(\alpha)$.

Threshold is arbitrary; common values are $p < .05$ or $p < .01$.

When is a finding “statistically significant”?



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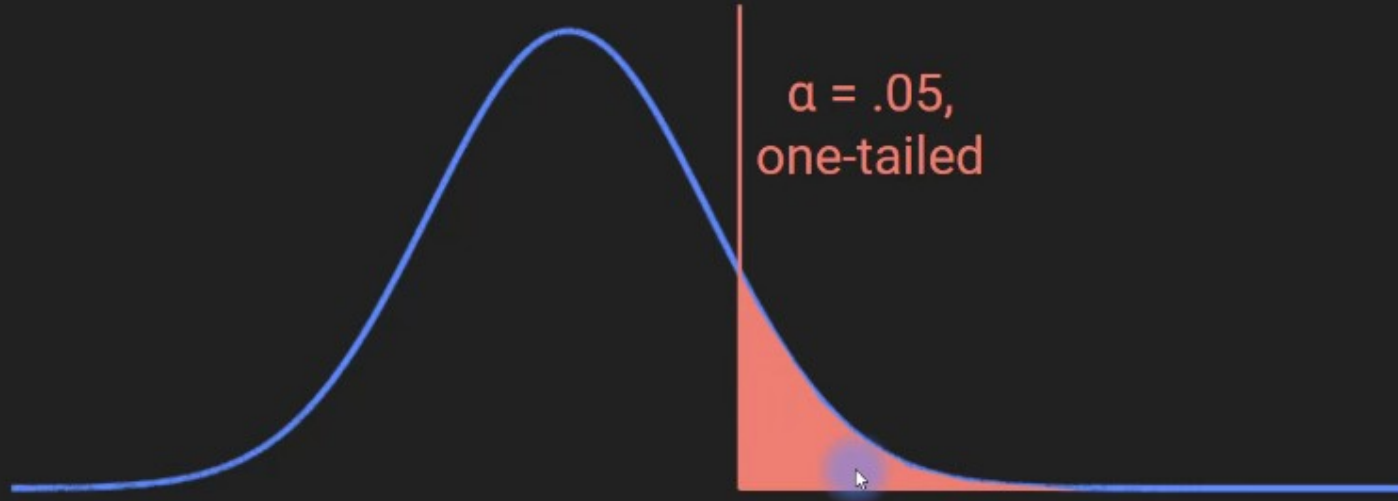
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Threshold is arbitrary; common values are $p < .05$ or $p < .01$.

"Tails" of a distribution



Each side of the H_0 distribution is unlikely.

The p-value threshold refers to the entire area of significance.

"Tails" of a distribution



Hypotheses should be as specific as possible (one-tailed).
But statistical tests are nearly always done two-tailed.

Here's what a p-value actually is. Read this slowly:

The p-value is the probability of getting your results (or something even MORE extreme) IF the null hypothesis were true.

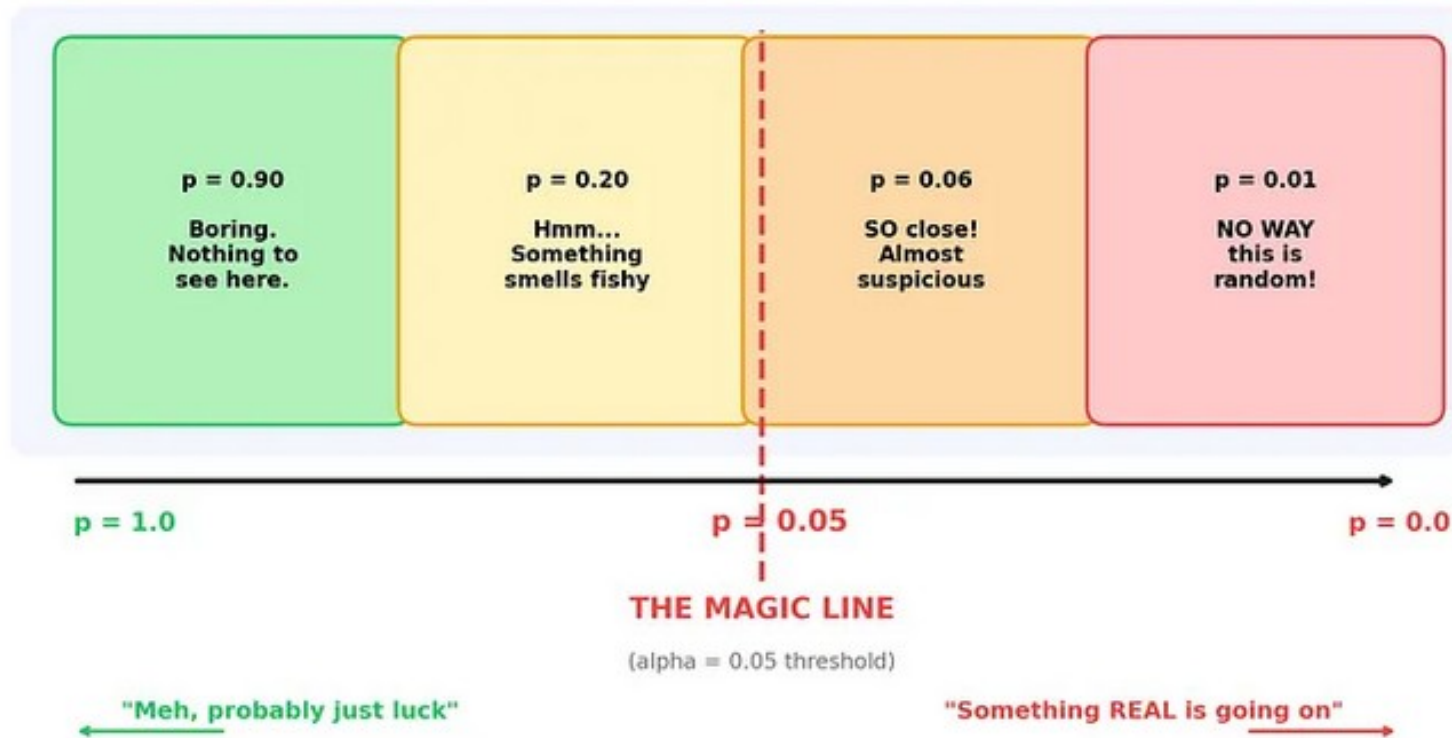
Let me unpack that, because every word matters.

The Scale of Suspicion

Think of the p-value as a **suspicion meter**:

- $p = 0.90$: “Boring. Nothing to see here. Go home.” This is like flipping a coin, getting 5 heads out of 10, and acting shocked. Completely normal.
- $p = 0.20$: “Hmm, a little odd, but not enough to make a fuss.” Like noticing your Uber is 1 minute late. Could be traffic.
- $p = 0.06$: “SO close to suspicious. Almost there. Just barely not enough.” Like checking 3 bad restaurant reviews out of 200. Probably fine? Maybe?
- $p = 0.01$: “No. Way. This is not random.” Like your Uber showing up 20 minutes late five times in a row. Something is broken.

P-Value: The Suspicion Meter



Common misinterpretations of p-values

Incorrect:

“My p-value is .02, so the effect is present for 2% of the population.”

Correct:

“My p-value is .02, so there is a 2% chance that there is no effect and my large sample statistic was due to sampling variability, noise, small sample size, or systematic bias.”

Common misinterpretations of p-values

Incorrect:

“My p-value is .02, so there is a 98% chance that my sample statistic equals the population parameter.”

Correct:

“My p-value is .02, so there is a 2% chance that there is no effect and my large sample statistic was due to sampling variability, noise, small sample size, or systematic bias.”

Common misinterpretations of p-values

Incorrect:

“My p-value is larger than the threshold, so therefore the null hypothesis is true.”

Correct:

“My p-value is larger than the threshold, so it is *likely* that the effect in the sample would have been observed given the null hypothesis, assuming that the sample is representative of the population. There could be many other explanations for the p-value other than H_0 .”

How to compute a p-value?

There are several ways to compute a p-value, and they depend on the specific statistical test and assumptions made.

You will learn specific methods later in the course.

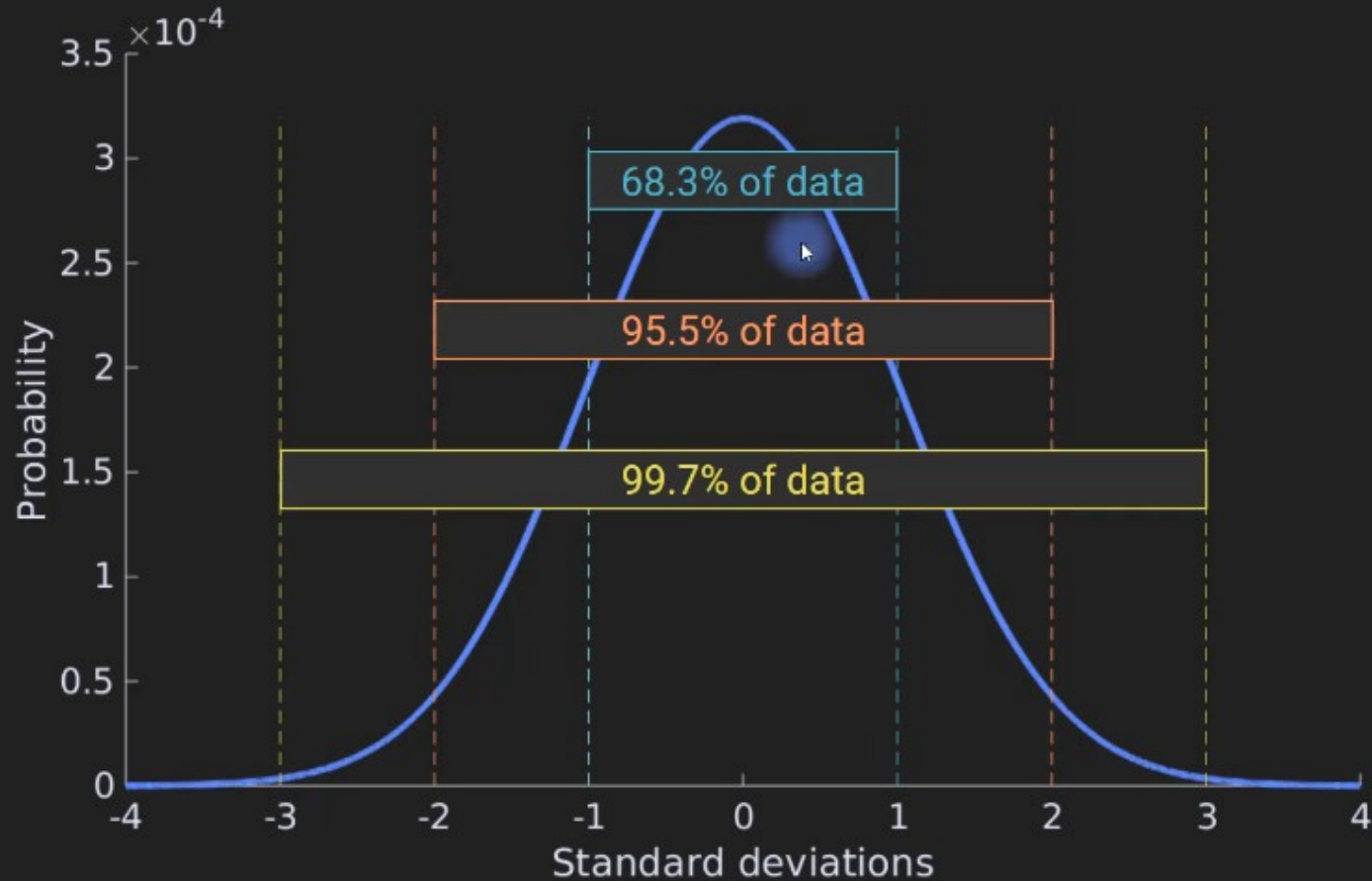
Importantly, the interpretation of a p-value is always the same, regardless of how the p-value was computed.

P-Z combinations that you should memorize

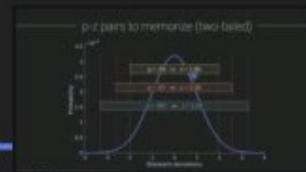
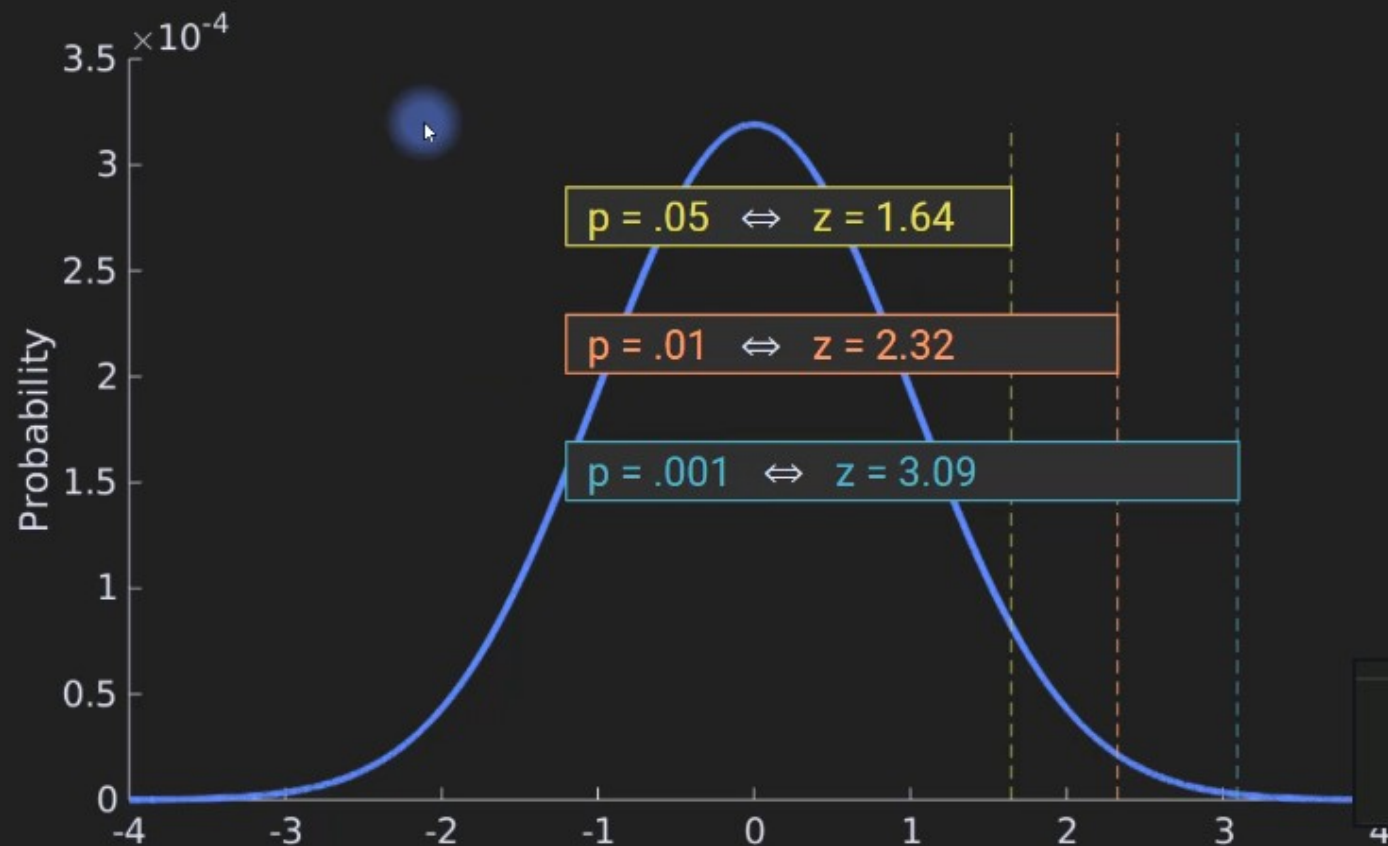
In this video, you will learn

- Z-data proportion values to memorize.
- P-value-standard deviation pairs to memorize.

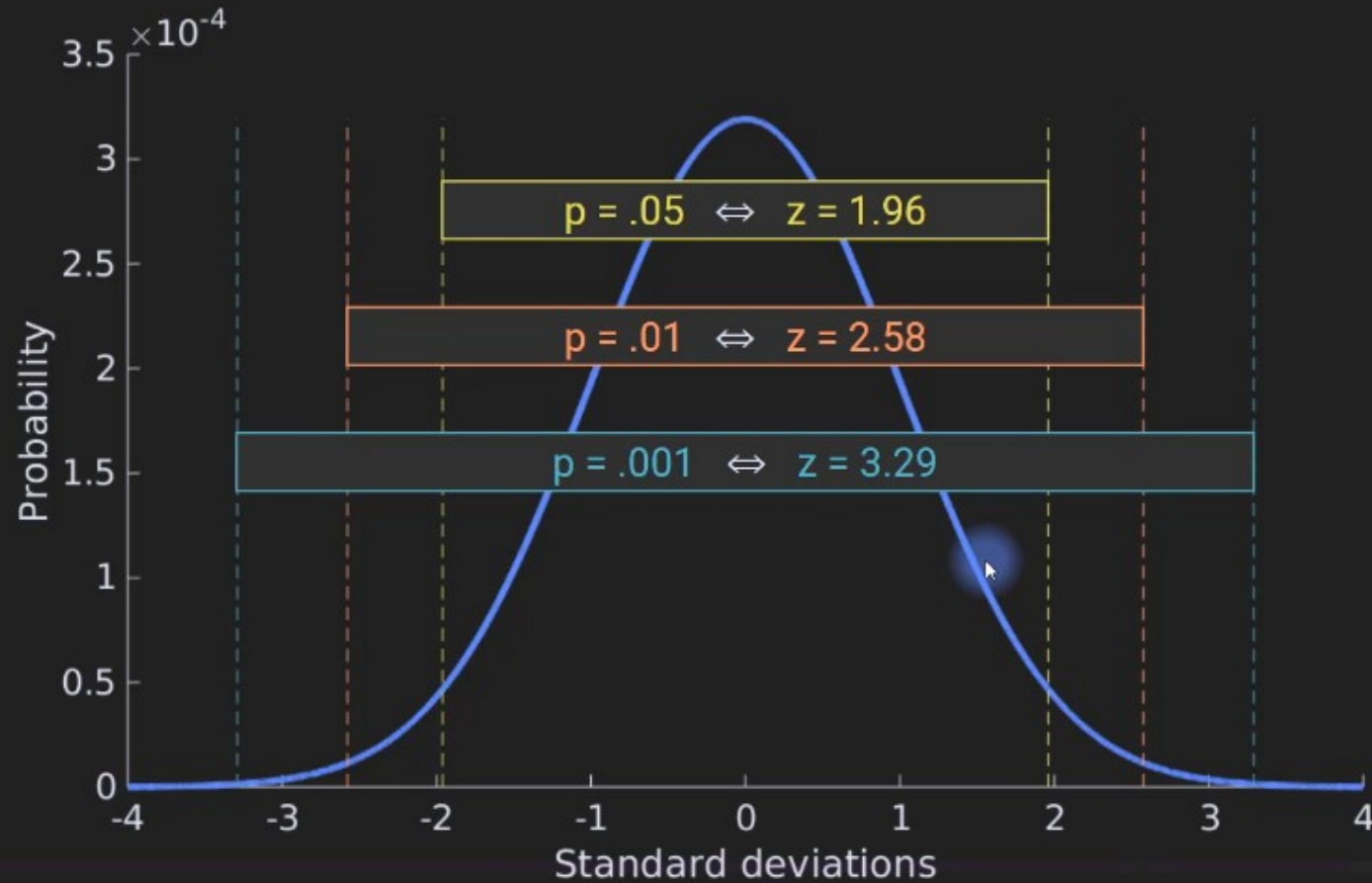
Z-proportion values to memorize



p-z pairs to memorize (one-tailed)



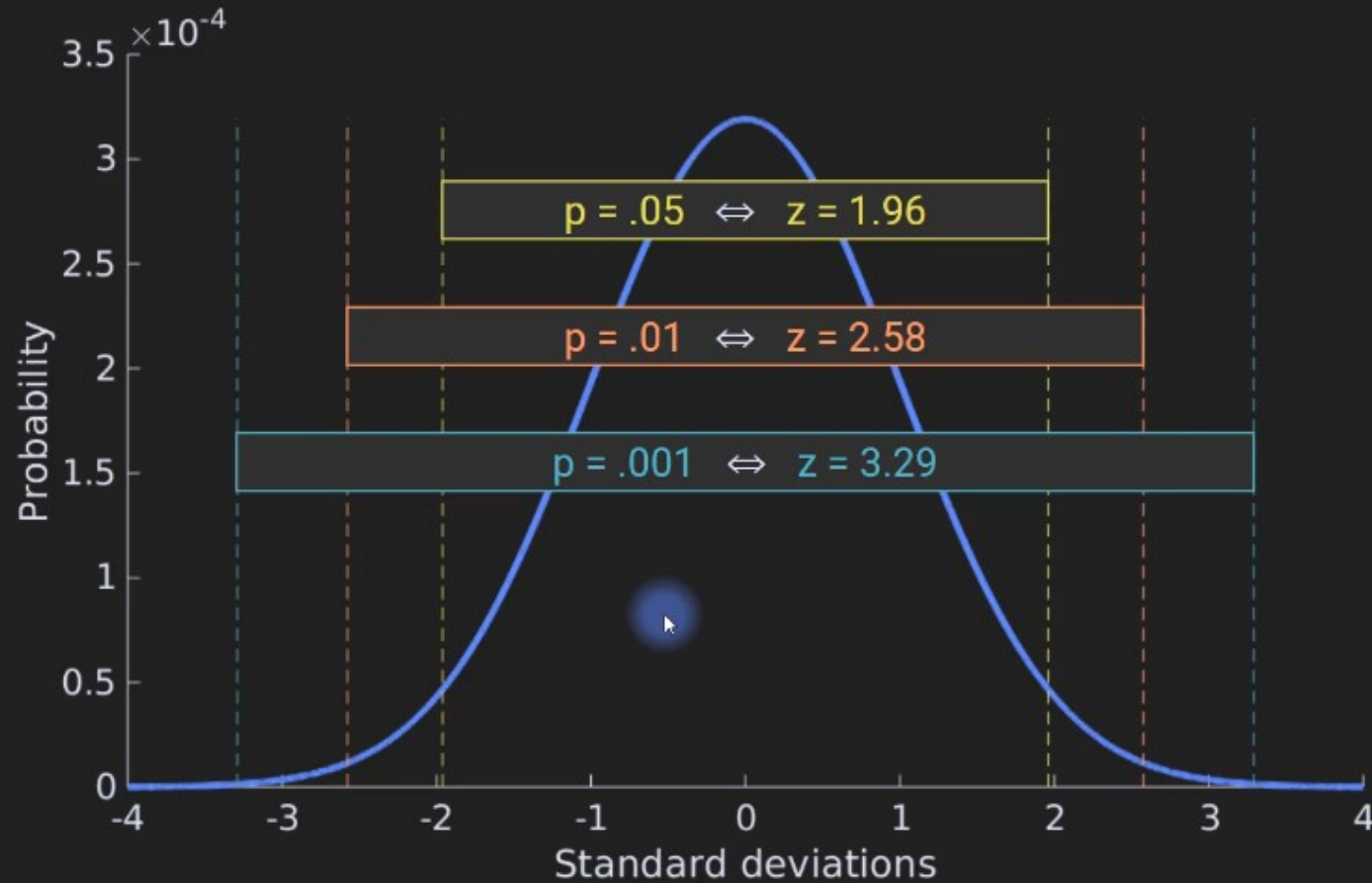
p-z pairs to memorize (two-tailed)



Örnek Senaryo:

Bir web sitesinin gecikme değerleri; $P50 = 50\text{ms}$, $P95 = 100\text{ms}$, $P99 = 500\text{ms}$ ise;
Kullanıcıların yarısı 50ms veya daha hızlı ($P50$),
Kullanıcıların %95'i 100ms veya daha hızlı ($P95$),
En yavaş %1'lik dilim ise 500ms 'den daha yavaş ($P99$) deneyim yaşıyor demektir.

p-z pairs to memorize (two-tailed)



Degrees of freedom

In this video, you will learn

- Several explanations for what “degrees of freedom” (df) means.

Degrees of freedom: numeric explanation

$$x = \{a, b, c, d\}$$

$$\bar{x} = 5$$

$$5 = \frac{a + b + c + d}{4}$$

$$x = \{3, 4, 9, d\}$$

$$d = 4$$

Degrees of freedom: numeric explanation

$$x = \{a, b, c, d\}$$

$$\bar{x} = 5$$

$$5 = \frac{a + b + c + d}{4}$$

$$x = \{3, 4, 9, d\}$$

$$d = \bar{x}n - a - b - c$$

Degrees of freedom: numeric explanation

$$x = \{a, b, c, d\}$$

$$\bar{x} = 5$$

This analysis has 3 degrees of freedom.

$$5 = \frac{a + b + c + d}{4}$$

We know one outcome variable and there are four samples.

$$x = \{3, 4, 9, d\}$$

$$d = \bar{x}n - a - b - c$$

Any three samples are unknown; the fourth is necessarily known.

Degrees of freedom: numeric explanation

$$x = \{a, b, c, d\}$$

$$\mu = 5$$

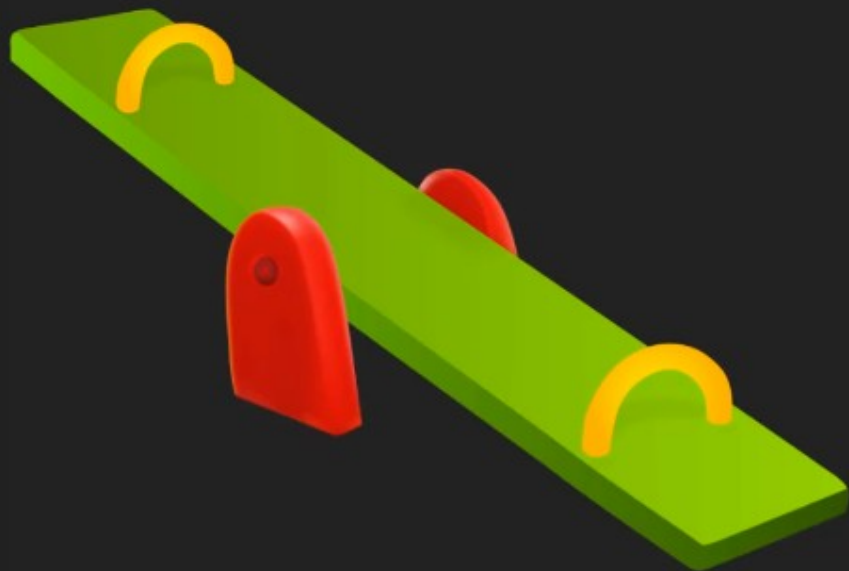
This analysis has 4 degrees of freedom.

The sample mean is not the same thing as the population mean.

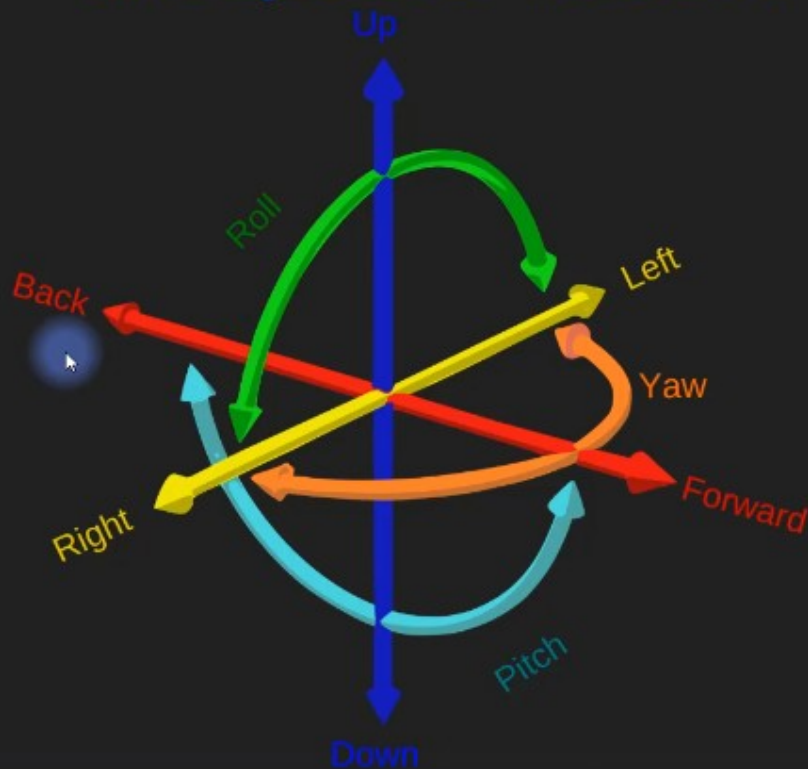
The population parameter does not constrain the sample data values.

Degrees of freedom: mechanical explanation

One degree of freedom



Six degrees of freedom



Why are degrees of freedom important?

Degrees of freedom (df) determine the shape of H_0 distributions.

Higher df generally indicates more power to reject the null hypothesis (related to statistical power).

df is also a way of checking an ANOVA table for accuracy and understanding.

Degrees of freedom definition

Degrees of freedom:

- The number of independent sample values.
- The number of sample data points that can vary.
- The number of data values that are unconstrained by the rest of the data values.

Degrees of freedom definition

Degrees of freedom:

- The number of independent sample values.
- The number of sample data points that can vary.
- The number of data values that are unconstrained by the rest of the data values.

Generally: $df = N - k$ (N data points, k parameters)

Type I and Type II errors

In this video, you will learn

- The mistakes of statistical significance labeling.
- How to minimize the chance of making these kinds of statistical errors.

Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Don't
reject H_0

Reject H_0

	H_0 is true	H_0 is false
Don't reject H_0		
Reject H_0		True positive ($1-\beta$)

H_0 distribution

H_A distribution

Significance
threshold (α)

Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Don't
reject H_0

True
negative
($1-\alpha$)

Reject H_0

True
positive
($1-\beta$)

H_0 distribution

H_A distribution

Significance
threshold (α)

Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Don't
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True
negative
($1-\alpha$)

Type II
(false neg.)
(β)

Reject H_0

True
positive
($1-\beta$)

H_0 distribution

Significance
threshold (α)

H_A distribution



Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Your decision

Don't
reject H_0

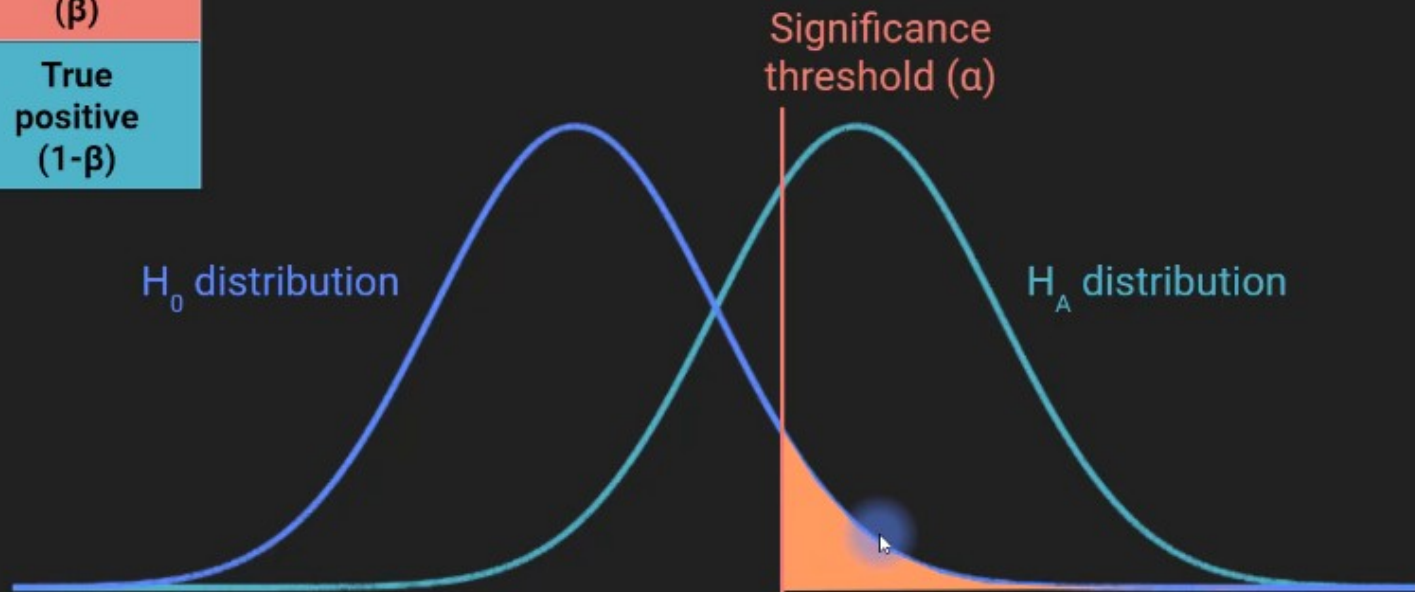
True
negative
($1-\alpha$)

Type II
(false neg.)
(β)

Reject H_0

Type I
(false pos.)
(α)

True
positive
($1-\beta$)



Statistical decisions

Reality about H_0

H_0 is true H_0 is false

Don't
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True
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positive
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H_0 distribution

H_A distribution

Significance
threshold (α)

Statistical trade-offs

Reality about H_0

H_0 is true H_0 is false

Your decision

Don't
reject H_0

True
negative
($1-\alpha$)

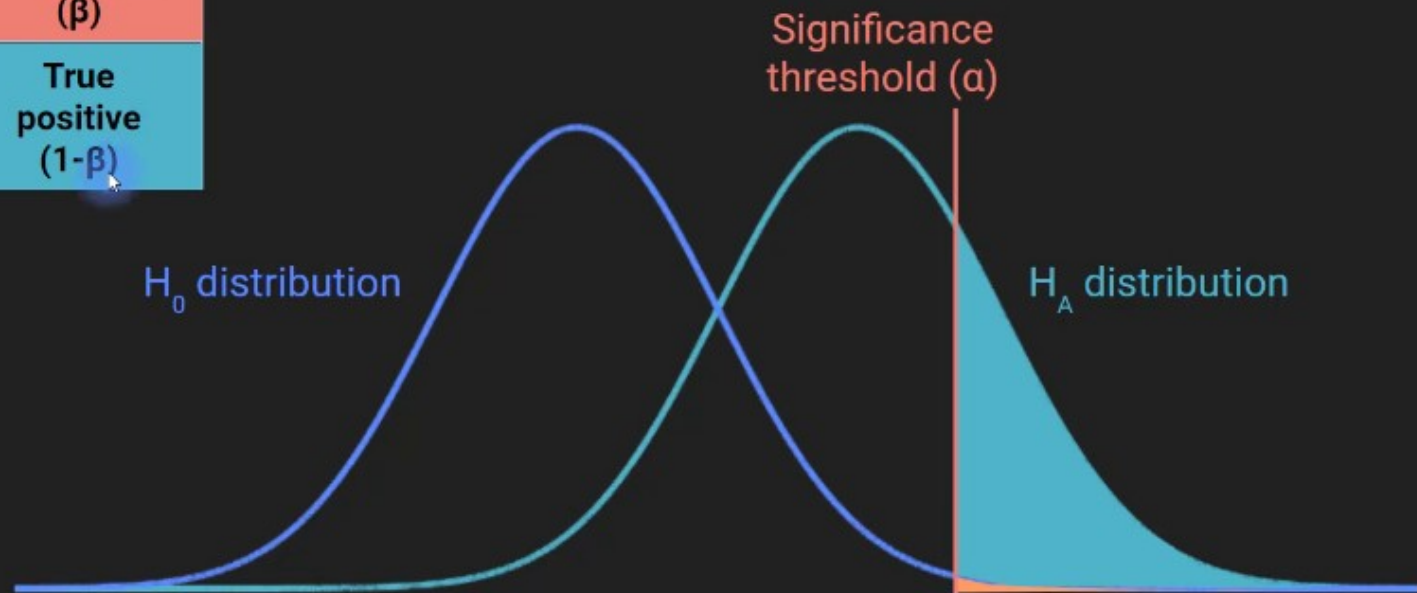
Type II
(false neg.)
(β)

Reject H_0

Type I
(false pos.)
(α)

True
positive
($1-\beta$)

Reducing **Type I errors** will
also reduce **True Positives**



Statistical trade-offs

Reality about H_0

H_0 is true H_0 is false

Your decision

Don't
reject H_0

True
negative
($1-\alpha$)

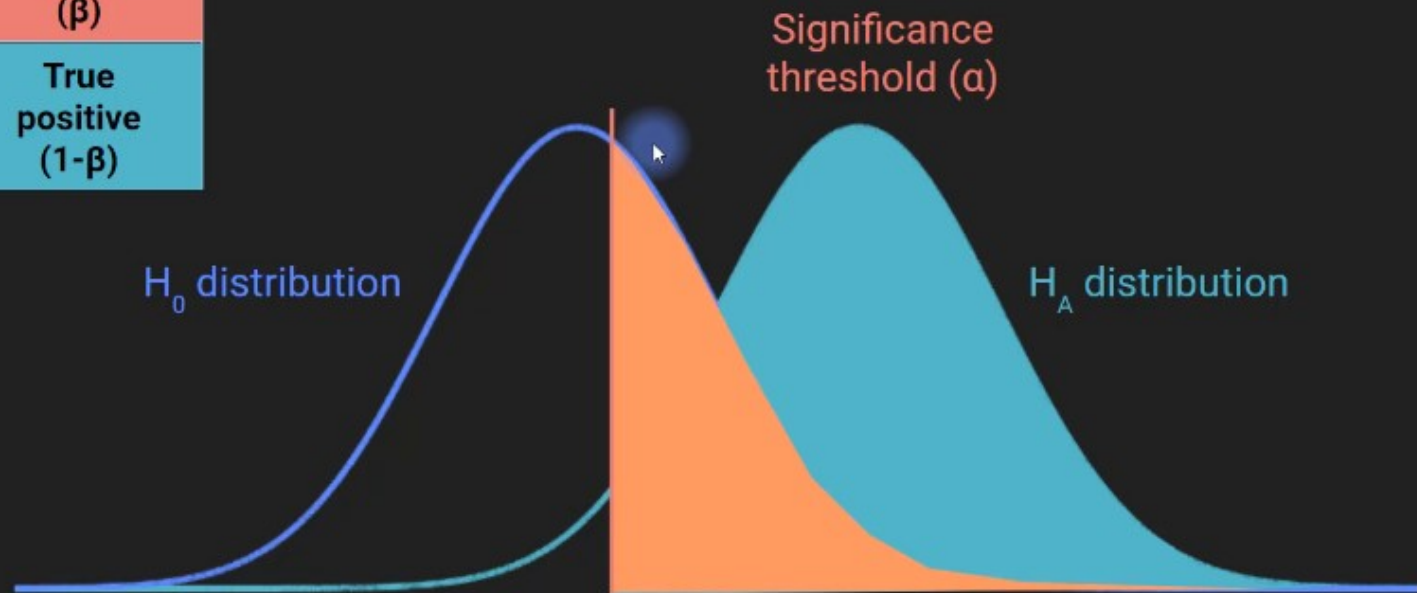
Type II
(false neg.)
(β)

Reject H_0

Type I
(false pos.)
(α)

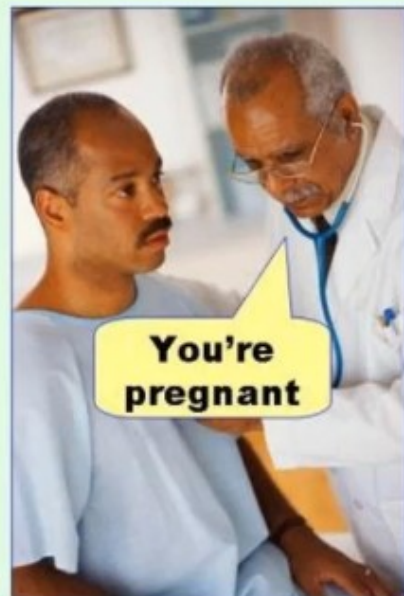
True
positive
($1-\beta$)

Increasing True Positives will
also increase Type I errors



Statistical errors in the real world (sort of)

Type I error
(false positive)



Type II error
(false negative)

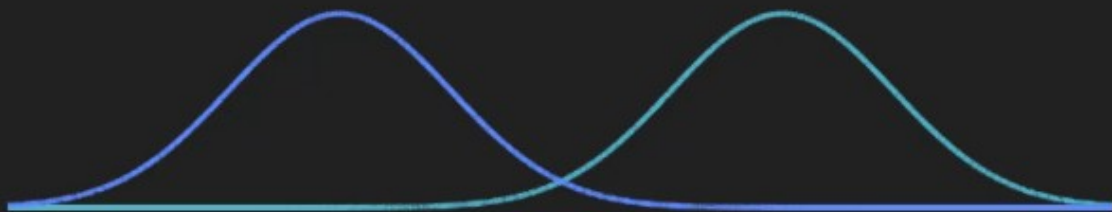


http://www.sethspielman.org/courses/geog5023/r_examples/Type_I_and_II_Errors.html

How to minimize statistical errors



To minimize statistical errors, you need to:



Increase the distance between the distributions (bigger effects).



Decrease the width of the distributions (less variable data).

Parametric vs. nonparametric statistics

In this video, you will learn

- The definition of nonparametric statistics.
- Advantages and limitations of each “family” of statistics.
- When to prefer nonparametric statistics.

Nonparametric statistics: definitions

What “non-parametric” *doesn't* mean:
No parameters at all.

Correct meaning(s):

Statistics that are not based on assumptions about underlying distributions (typically, Gaussian).

Statistical inference methods that generate the H_0 distribution from the data, not from an equation.

Statistical model complements

Parametric test

1-sample t-test

2-sample t-test

Pearson correlation

ANOVA

Nonparametric test

Wilcoxon sign-rank test

Mann-Whitney U test

Spearman correlation

Kruskal-Wallis test

Important application of nonparametric statistics:
Permutation testing and cross-validation

Advantages and limitations

Parametric statistics:

- Standard, widely used
- Based on assumptions
- Assumptions should be tested, (though rarely done)
- Can be incorrect when assumptions are violated
- Computationally fast
- Analytically proven

Nonparametric statistics:

- Some are nonstandard
- “No” assumptions necessary
- Can be slow/intensive
- Some are sensible algorithms rather than proven methods
- Appropriate for non-numeric data
- Appropriate for small sample sizes
- Some methods give different results each time

Conclusion:

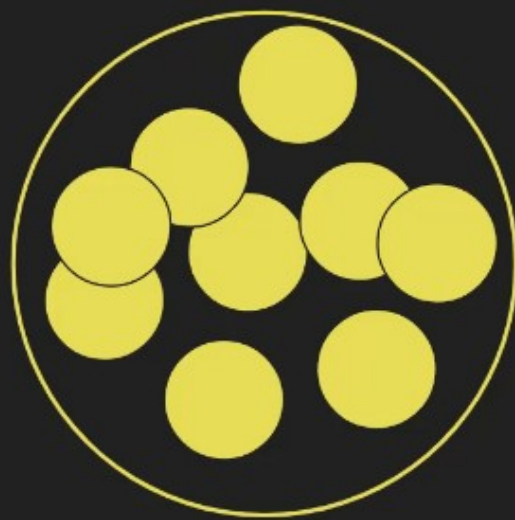
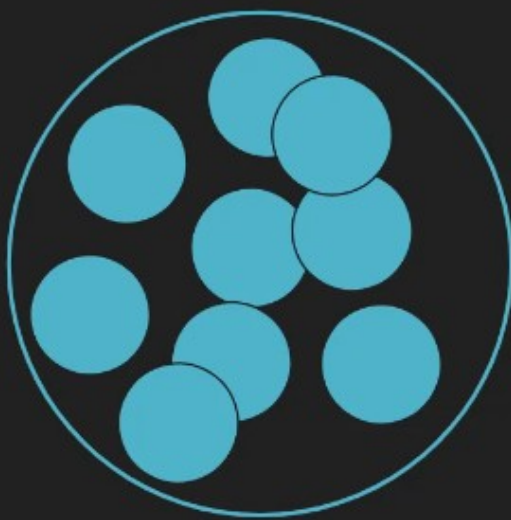
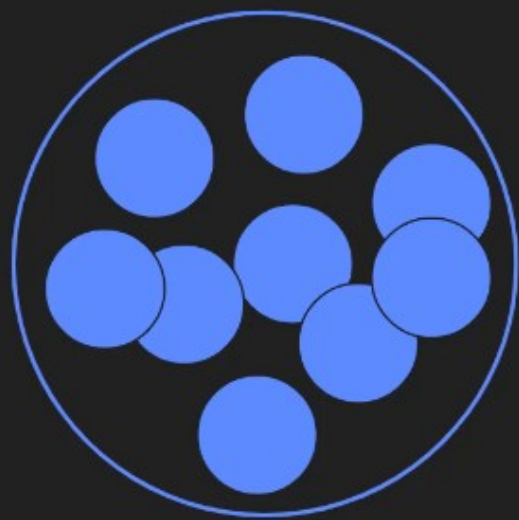
Use parametric methods when possible.
Use nonparametric methods when necessary.

Multiple comparisons and Bonferroni correction

In this video, you will learn

- What “multiple comparisons” means and why it causes problems.
- How Bonferroni correction works.
- The difference between individual Type 1 error rate and familywise error rate.

Mo' groups, mo' comparisons



$H_1 =$  vs. 

$H_2 =$  vs. 

$H_3 =$  vs. 

Mo' comparisons, mo' problems

Probabilities can be additive.

$$p(H_1 | H_0) = .05$$

$$p(H_2 | H_0) = .05$$

$$p(H_3 | H_0) = .05$$

$$p(H_1 | H_0) + p(H_2 | H_0) + p(H_3 | H_0) = .05 + .05 + .05 = .15$$

Problem: The probability of a false alarm
(Type I error) is 15%! Unacceptably high!

Mo' comparisons, mo' problems

Problem:

The true “familywise error rate” (FWE or FWER) is impossible to calculate exactly, because of possible dependencies between hypotheses.

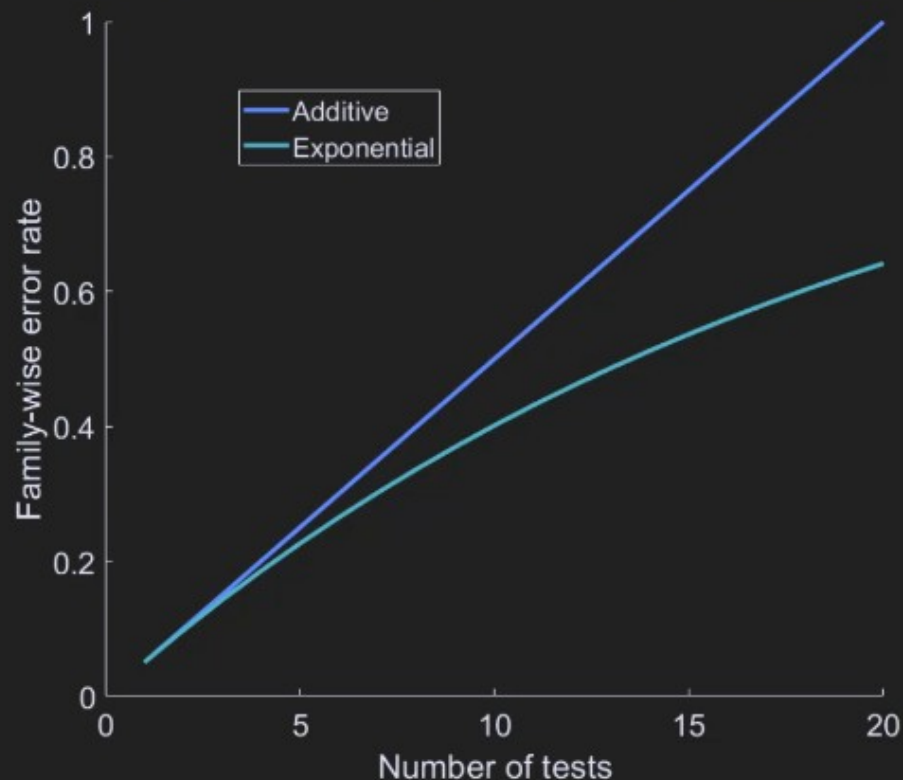
Solution:

Estimate the maximum FWE.

$$\text{FWE} \leq 1 - (1 - \alpha)^n \leq n\alpha$$

Mo' comparisons, mo' problems

$$\text{FWE} \leq 1 - (1 - \alpha)^n \leq n\alpha$$



The Italians come to the rescue!



Bonferroni correction:

$$\text{Threshold} = \alpha / N$$

α = e.g., 0.05

N = number of tests

Mo' comparisons, mo' problems

Assumption: probabilities are additive.

$$p(H_1 | H_0) = .05/3$$

$$p(H_2 | H_0) = .05/3$$

$$p(H_3 | H_0) = .05/3$$

$$p(H_1 | H_0) + p(H_2 | H_0) + p(H_3 | H_0) = .05/3 + .05/3 + .05/3 = .15/3 = .05$$

Note the difference between the individual α ($p=.0167$) and the familywise α ($p=.05$).

Statistical vs. theoretical vs. clinical significance

In this video, you will learn

- The differences among the different “significances.”
- Why $p < .05$ is not the only measure of importance.

The significances

Statistical significance:

The probability of observing a test statistic this large given that the null hypothesis is true.

Theoretical significance:

A finding is relevant for a theory or leads to new experiments. This has nothing to do with statistical significance.

Clinical (practical, societal, educational) significance:

A finding is relevant for diagnosing or treating a disease.

Examples

Hypothesis and result:

MMR vaccine causes autism. $p = .79$.

No statistical significance.

Strong clinical and societal significance.

Examples

Hypothesis and result:

Piles of sand with larger grains collapse sooner. $p = .001$.

Strong statistical significance.

Strong theoretical significance (scale-free dynamics)

No clinical or societal significance.

Examples

Hypothesis and result:

People who take Mike X Cohen's course get better grades in their stats courses. $p < 10^{-30}$.

Strong statistical significance.

No theoretical significance.

Some societal significance?

No clinical significance.

Strong individual practical significance.

Cross-validation

In this video, you will learn

- What “cross-validation” means.
- The terms “training” and “testing” data.
- Assumptions of cross-validation.

Cross-validation

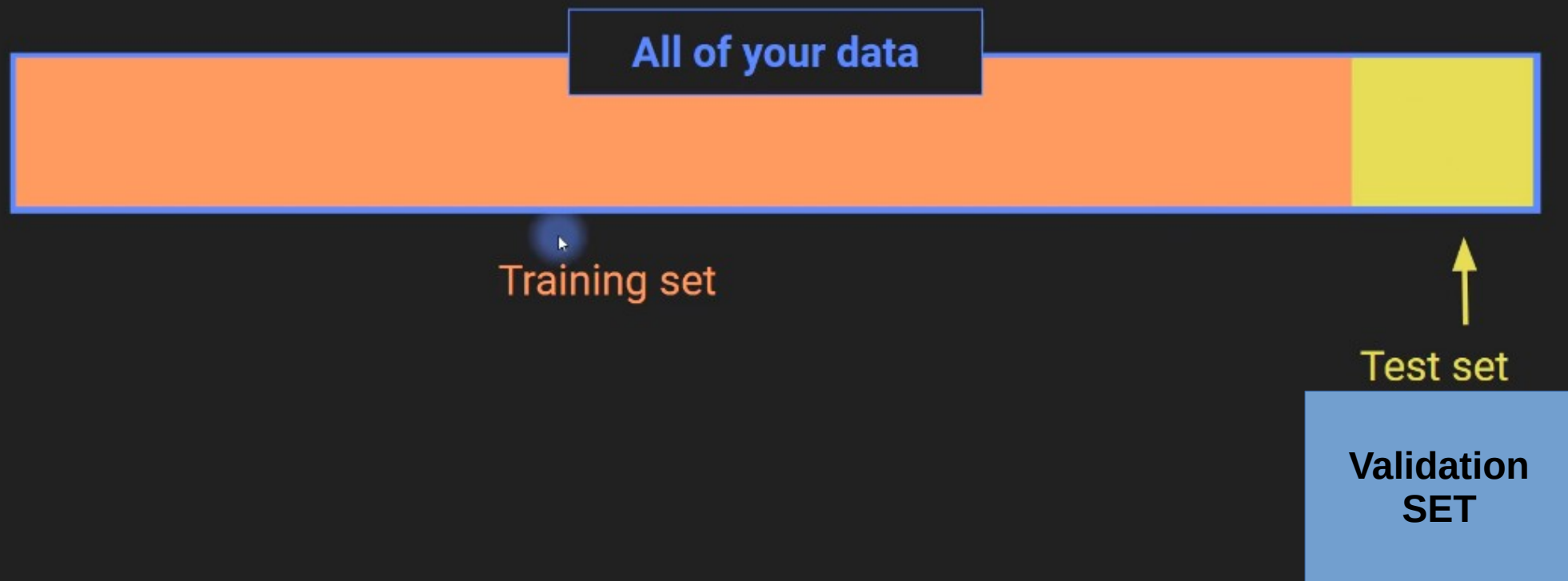
All of your data

A horizontal bar representing the entire dataset. It is divided into two sections: a large orange section on the left and a smaller yellow section on the right. A black box with the text 'All of your data' is positioned above the orange section. Below the orange section is the text 'Construct the statistical model on these data'. Below the yellow section is an upward-pointing arrow with the text 'Apply the model to these data'.

Construct the statistical model on these data

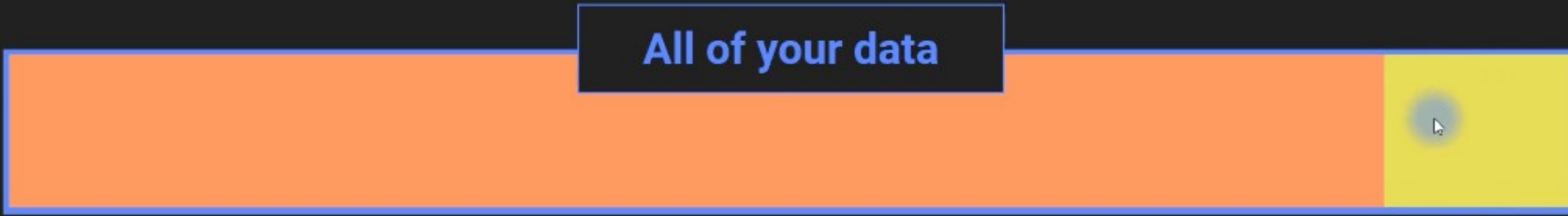
Apply the model to these data

Cross-validation



Cross-validation

All of your data

A horizontal bar representing the entire dataset. The bar is divided into two sections: a large orange section on the left and a smaller yellow section on the right. A black box with the text 'All of your data' is positioned above the orange section. A yellow arrow points upwards from the text 'Out-of-sample data' to the yellow section. The entire bar is outlined in blue.

In-sample data

Out-of-sample data

Cross-validation



A horizontal bar representing the entire dataset is divided into three segments. The first segment on the left is orange, the middle segment is yellow, and the third segment on the right is orange. A black box with the text "All of your data" is positioned above the yellow segment. A mouse cursor is pointing at the yellow segment.

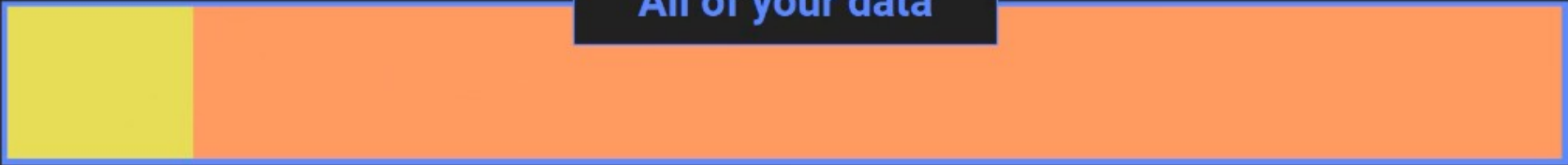
All of your data

“K-fold cross validation”

What is an appropriate k? Usually 10 or 20.

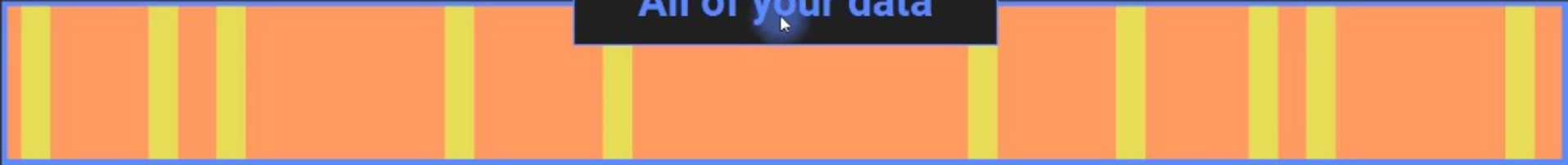
How to separate training and testing data

All of your data



A horizontal bar representing the entire dataset. The bar is mostly orange, with a small yellow segment at the left end. A black box with the text "All of your data" is positioned above the orange portion of the bar.

All of your data



A horizontal bar representing the entire dataset, divided into many small segments. Most segments are orange, while a few are yellow. A black box with the text "All of your data" is positioned above the bar.

Uses of cross-validation

Variance of an estimate (e.g., confidence intervals)

Also called “jack-knifing.” Compare the variance of the parameter estimates over different training sets.

Avoid bias in analysis results

Applying the model to data it has never seen. Overfitting the training data will decrease performance on the test data.

Compute classification accuracy

Dominant application in machine learning and deep learning.

Uses of cross-validation

Ideally, the test set is truly independent of the training set.
This avoids bias and overfitting

Be mindful of whether this is the case in your data.
For example, a group of self-selected students from the same school.

Statistical significance vs. classification accuracy

In this video, you will learn

- How models classify data.
- The key differences between classification and p-values.

How classification works

We want to predict whether a student will pass or fail the statistics course.

The model uses study hours, alcoholic drinks, and web-surfing hours to predict this outcome.

How classification works

We input the students' behavior (study, drinking, webbing) into the model, and the model gives an output.

The model output is not a label; it is a probability, e.g., pass 23% / fail 77%.

Model output is thresholded at 50% to make a binary prediction. This is called *classification*.

Data types for classification

Classification is mainly appropriate for categorical data (nominal and ordinal).

Discrete is also possible.

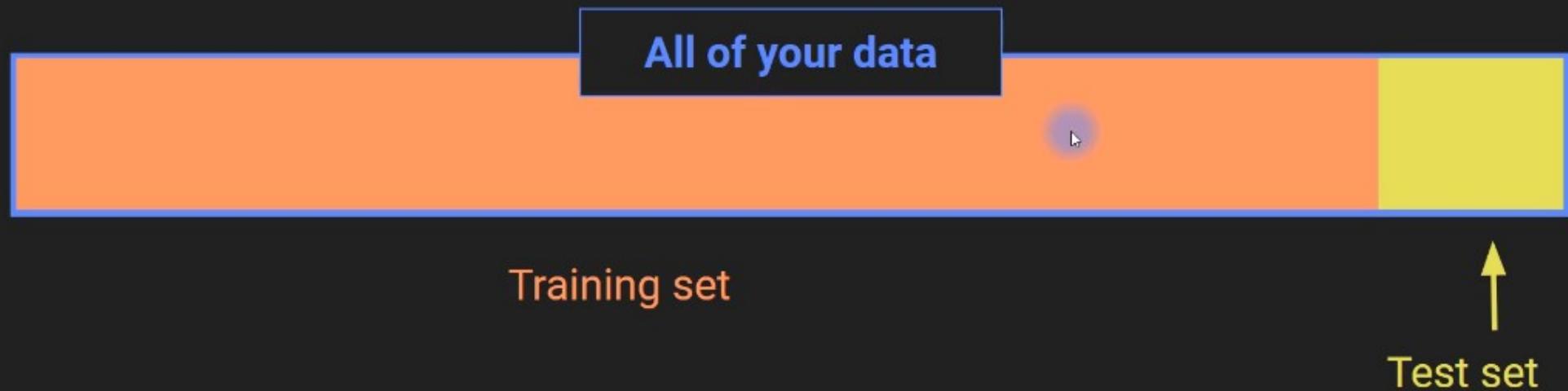
Why not interval or ratio data?

How classification works

The model prediction is compared to the reality, averaged, and called classification accuracy.

Model output	Predicted category	Reality	Accuracy
.77	FAIL	FAIL	1
.15	PASS	PASS	1
.92	FAIL	FAIL	1
.56	FAIL	PASS	0

Classification and cross-validation



Model accuracy vs. p-value

P-value:

- Tests the probability of the sample statistic (or model) by chance.
- Can be computed separately for each parameter in the model.
- Analytic solutions with established mathematical theory.
- Can be obtained for nearly any kind of model with any kind of variables.
- Sensitive to extreme N's.

Classification accuracy:

- Tests how closely the model output matches an observed outcome.
- Individual parameter contributions can be difficult to determine.
- Sensible empirical method. Results may differ for each run.
- Used only for certain kinds of models and variables.
- Robust to sample size.

Model accuracy vs. p-value

P-value:

- Tests the probability of the sample statistic (observed) occurring by chance.
- Can be computed for each parameter.
- Analytic solution using mathematical formulas.
- Can be obtained for each of model variables.
- Sensitive to extreme N's.

Classification accuracy:

- Tests how closely the model output matches the desired outcome.
- Contributions of each feature to the model outcome are determined.
- Results are more intuitive.
- Can be used for a variety of kinds of models.
- Robust to sample size.

IMPORTANT: It's not a competition; classification and p-values are different evaluations of a model. They can both be used without conflicting, interfering, or misleading.